

BREMGARTEN, GermanyWMO: **109000**Lat: **47.900N**Lon: **7.617E**Elev: **217**StdP: **98.75**Time Zone: **1.00 (EUC)**Period: **82-92**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.6	-8.2	-13.9	1.2	-11.1	-10.4	1.6	-7.2	13.6	8.0	12.4	7.7	1.3	200	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	32.1	20.9	30.0	19.9	28.2	19.0	21.6	29.2	20.7	28.3	19.7	26.9	3.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.0	14.2	25.1	18.0	13.3	24.1	17.1	12.5	23.2	63.7	29.7	60.4	28.1	57.4	26.9	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.1	9.6	8.5	-13.6	36.4	5.2	1.5	-17.3	37.5	-20.4	38.4	-23.3	39.2	-27.1	40.3		
			-13.7	23.8	5.3	1.1	-17.5	24.5	-20.6	25.2	-23.5	25.8	-27.4	26.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	1.5	1.9	6.7	9.8	14.3	17.7	20.7	20.1	16.6	11.7	5.3	3.0	(6)
(7)		DBStd	7.86	5.47	5.18	3.82	3.40	3.69	3.32	3.26	2.88	3.33	3.70	4.29	4.56	(7)
(8)		HDD10.0	1052	265	228	115	44	7	0	0	0	0	25	147	220	(8)
(9)		HDD18.3	2960	522	459	362	256	133	52	13	17	72	208	391	474	(9)
(10)		CDD10.0	1355	2	2	12	39	139	230	333	312	199	76	7	3	(10)
(11)		CDD18.3	221	0	0	0	0	7	32	88	71	21	2	0	0	(11)
(12)		CDH23.3	2374	0	1	3	11	116	338	954	721	207	23	0	0	(12)
(13)		CDH26.7	778	0	0	0	1	18	87	380	243	46	5	0	0	(13)
(14)	Wind	WSAvg	3.0	3.8	3.1	3.7	3.3	2.9	3.1	2.5	2.4	2.7	2.6	2.9	3.3	(14)
(15)	Precipitation	PrecAvg	947	58	58	60	71	99	95	93	88	80	83	81	79	(15)
(16)		PrecMax	1231	133	113	203	152	171	176	234	218	164	198	212	145	(16)
(17)		PrecMin	723	6	21	16	4	38	33	37	14	25	1	1	4	(17)
(18)		PrecStd	134	35	26	39	41	41	40	40	41	35	38	44	39	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	16.9	22.9	25.0	28.9	31.8	35.9	34.2	30.2	26.9	17.8	15.2	(19)
(20)			MCWB	9.6	11.1	13.3	14.5	18.1	21.1	21.8	20.0	19.4	17.5	12.4	10.8	(20)
(21)		2%	DB	11.9	13.6	17.5	22.0	26.2	29.0	33.0	31.2	28.0	21.9	15.1	13.6	(21)
(22)			MCWB	8.5	9.0	10.7	12.7	16.4	19.2	21.2	19.8	19.0	15.8	11.4	10.0	(22)
(23)		5%	DB	10.1	10.9	14.9	19.2	24.1	27.1	30.8	29.1	25.9	19.0	13.1	11.8	(23)
(24)			MCWB	7.4	7.8	9.5	11.8	16.0	18.6	20.5	19.4	18.1	14.2	10.3	8.9	(24)
(25)		10%	DB	8.8	9.0	12.9	17.0	21.9	25.0	28.8	27.2	23.8	17.0	11.2	9.9	(25)
(26)			MCWB	6.5	6.7	8.5	10.6	14.8	17.5	19.8	18.7	17.2	13.1	8.8	7.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	11.1	13.5	15.1	19.0	22.0	23.8	21.8	21.0	17.6	13.3	11.9	(27)
(28)			MCD B	13.7	15.8	21.7	23.4	26.1	29.4	32.4	30.5	28.4	24.5	16.2	14.1	(28)
(29)		2%	WB	8.7	9.6	11.2	13.5	17.3	20.2	22.3	20.8	19.7	16.2	11.8	10.1	(29)
(30)			MCD B	11.6	12.9	16.4	20.2	24.5	27.2	29.9	28.6	26.3	20.6	14.4	12.8	(30)
(31)		5%	WB	7.5	8.0	10.0	12.4	16.2	19.1	21.2	20.0	18.5	14.7	10.6	9.1	(31)
(32)			MCD B	9.8	10.4	13.7	18.0	22.4	26.1	28.8	27.8	24.4	17.8	13.1	11.7	(32)
(33)		10%	WB	6.3	6.6	8.9	11.2	15.3	17.9	20.2	19.3	17.4	13.6	9.2	7.5	(33)
(34)			MCD B	8.3	8.8	12.7	16.1	21.0	23.8	27.6	26.3	22.5	16.4	11.3	9.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.0	8.0	9.5	10.9	11.3	11.2	12.8	12.8	11.6	9.2	7.0	5.9	(35)
(36)			MCD BR	7.4	11.8	13.7	16.3	16.3	15.3	17.0	16.5	15.2	12.6	9.7	7.9	(36)
(37)			MCWBR	5.5	8.0	8.1	8.7	8.2	7.2	7.3	7.1	7.3	7.4	6.7	5.6	(37)
(38)		5% WB	MCD BR	7.1	10.2	11.8	14.7	13.9	14.1	14.9	14.7	13.7	11.6	9.3	7.3	(38)
(39)			MCWBR	5.4	7.3	7.3	8.0	7.4	6.9	6.8	6.6	7.1	7.0	6.8	5.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.315	0.341	0.374	0.411	0.421	0.426	0.424	0.419	0.394	0.371	0.336	0.312	(40)	
(41)		taud	2.439	2.360	2.297	2.215	2.236	2.260	2.260	2.291	2.336	2.386	2.436	2.437	(41)	
(42)		Ebn at Noon	760	812	837	838	844	840	835	823	807	765	723	714	(42)	
(43)		Edh at Noon	66	89	110	131	134	131	130	121	105	86	67	59	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.02	1.81	3.04	4.32	5.05	5.90	5.58	4.85	3.67	2.26	1.17	0.82	(44)	
(45)		RadStd	0.14	0.30	0.47	0.63	0.53	0.52	0.52	0.47	0.36	0.20	0.14	0.15	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (103 neighbors)	N/A	N/A	N/A	N/A	+0.39	+0.30	N/A	N/A	+52	+16

Nomenclature: See separate page